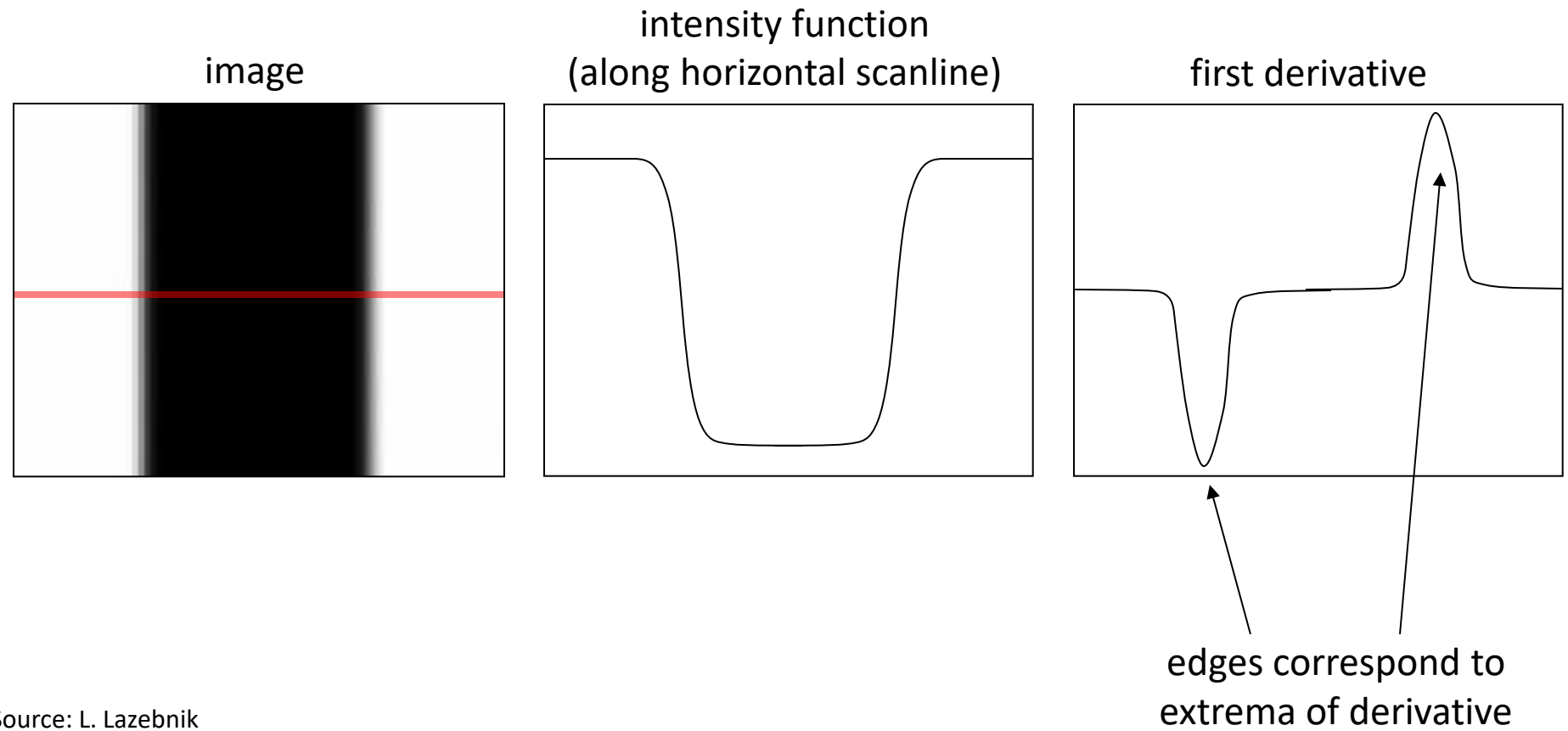
# Characterizing edges

- An edge is a place of *rapid change* in the image intensity function




Source: L. Lazebnik

# Image derivatives

- How can we differentiate a *digital* image F[x,y]?
  - Option 1: reconstruct a continuous image, *f,* then compute the derivative
  - Option 2: take discrete derivative (finite difference)

$$\frac{\partial f}{\partial x}[x, y] \approx F[x+1, y] - F[x, y]$$

How would you implement this as a linear filter?

$\frac{\partial f}{\partial x}$:

| | | |
|---|---|---|
| | | |
| | | |
| | | |

$H_x$

$\frac{\partial f}{\partial y}$:

| | | |
|---|---|---|
| | | |
| | | |
| | | |

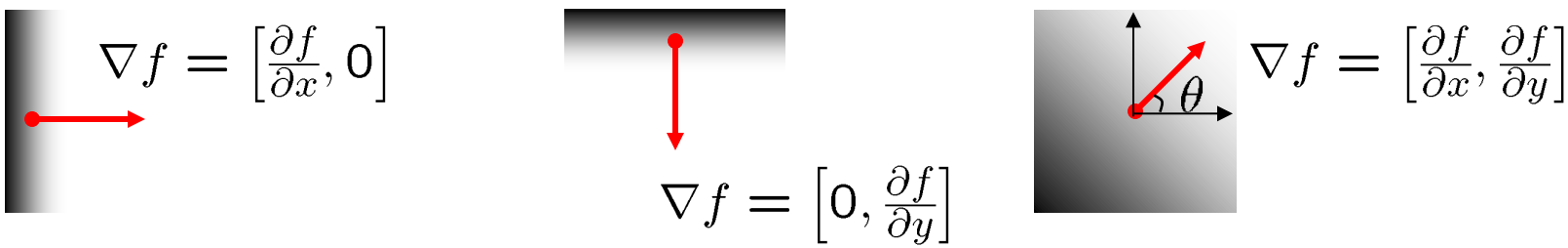
$H_y$

Source: S. Seitz

# Image gradient

- The *gradient* of an image: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

  The gradient points in the direction of most rapid increase in intensity




The *edge strength* is given by the gradient magnitude:

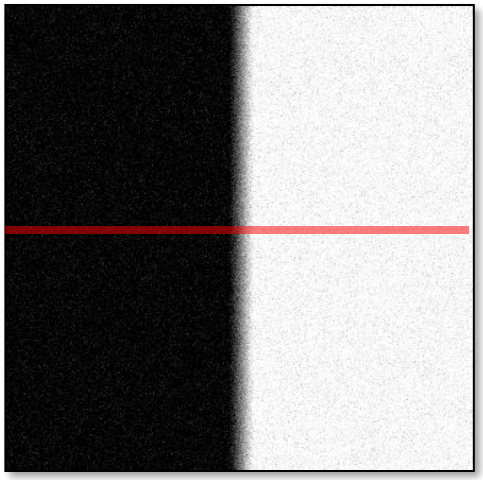
$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

The gradient direction is given by:

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

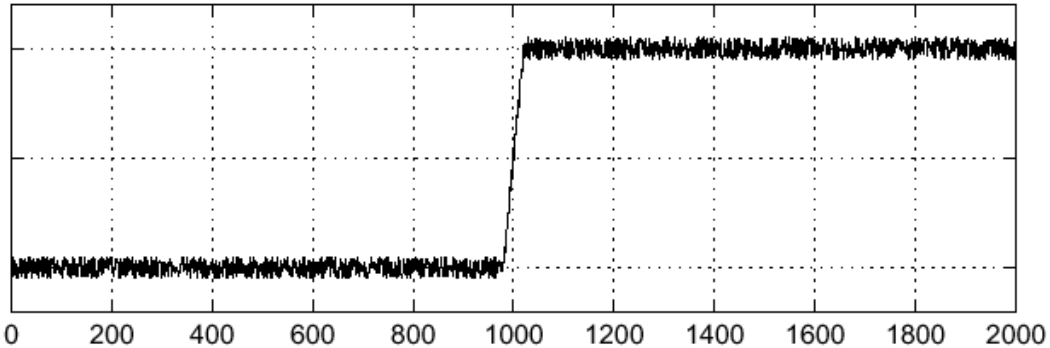
Source: Steve Seitz

# Effects of noise

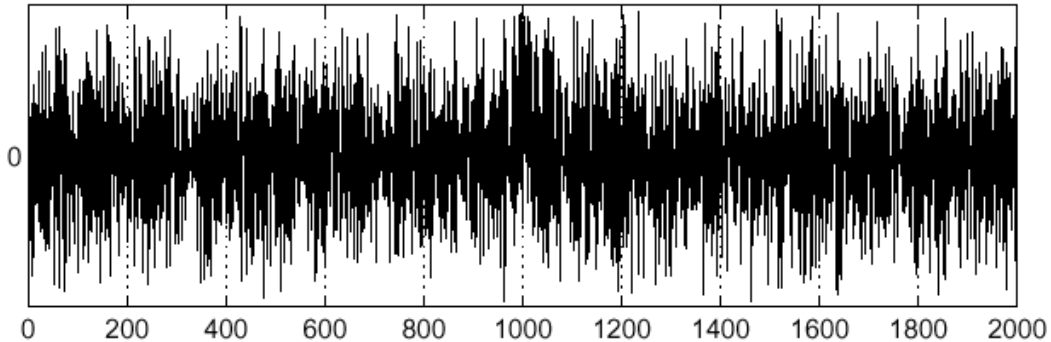


Noisy input image

$f(x)$



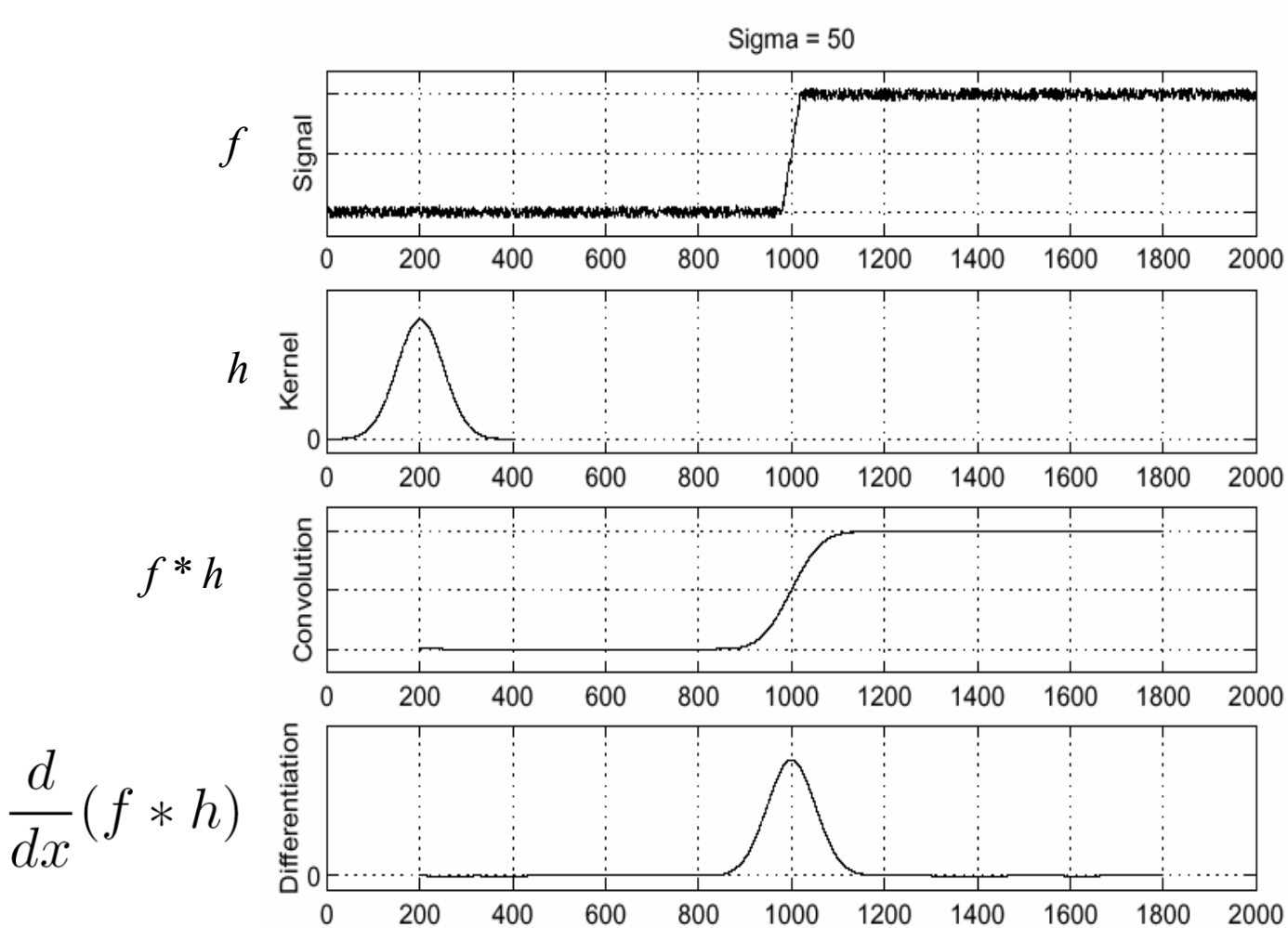

$\frac{d}{dx}f(x)$




Where is the edge?

Source: S. Seitz

# Solution: smooth first



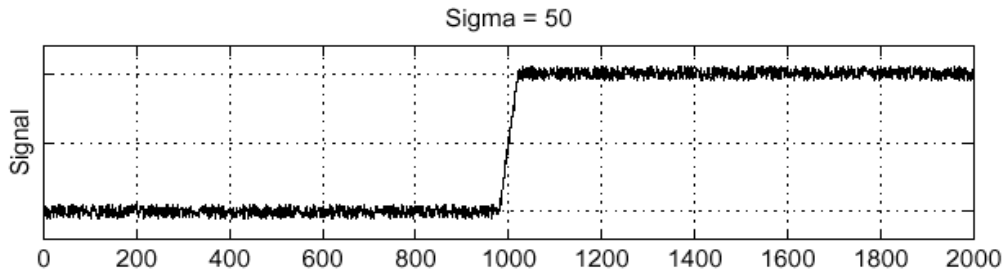

To find edges, look for peaks in $\frac{d}{dx}(f * h)$

Source: S. Seitz

# Associative property of convolution

- Differentiation is convolution, and convolution is associative: $\frac{d}{dx}(f * h) = f * \frac{d}{dx}h$
- This saves us one operation:

$f$

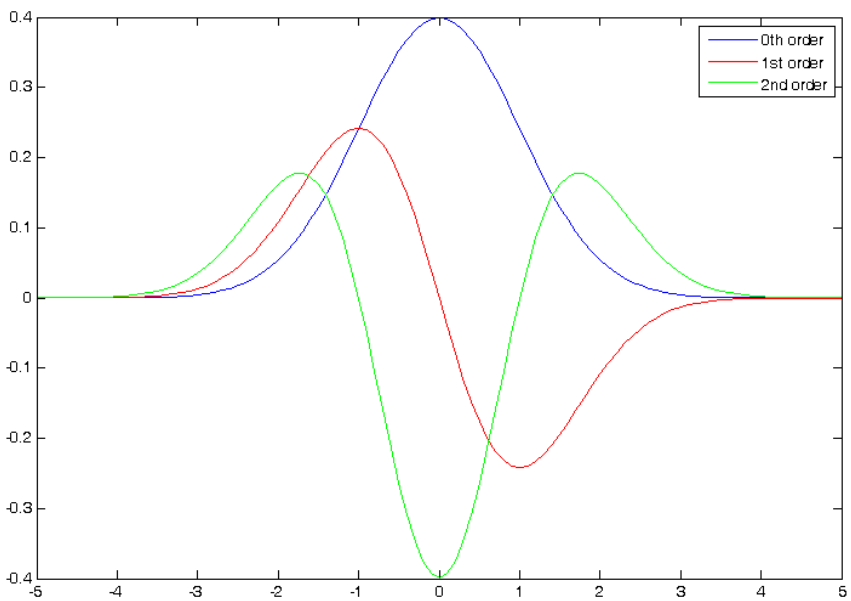



Source: S. Seitz

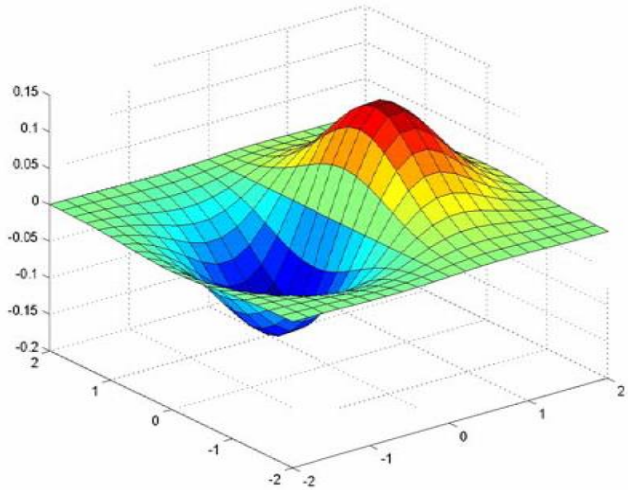
# The 1D Gaussian and its derivatives

$$G_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

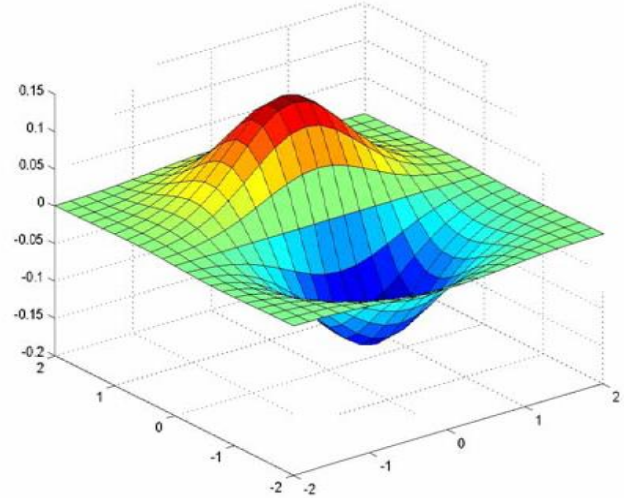
$$G'_\sigma(x) = \frac{d}{dx} G_\sigma(x) = -\frac{1}{\sigma}\left(\frac{x}{\sigma}\right) G_\sigma(x)$$



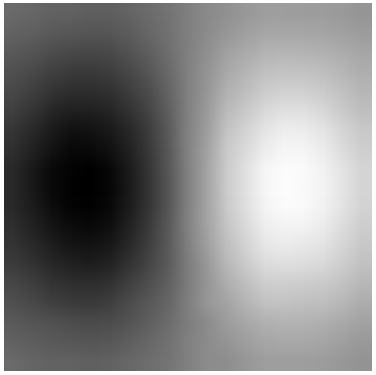
# Derivative of Gaussian filter

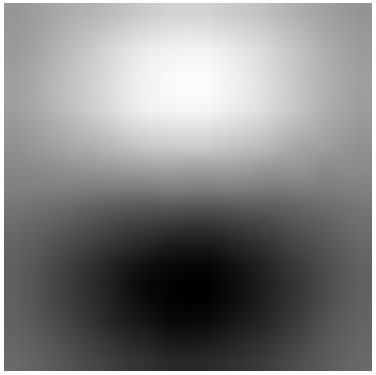



*x*-direction




*y*-direction





# The Sobel operator

- Common approximation of derivative of Gaussian

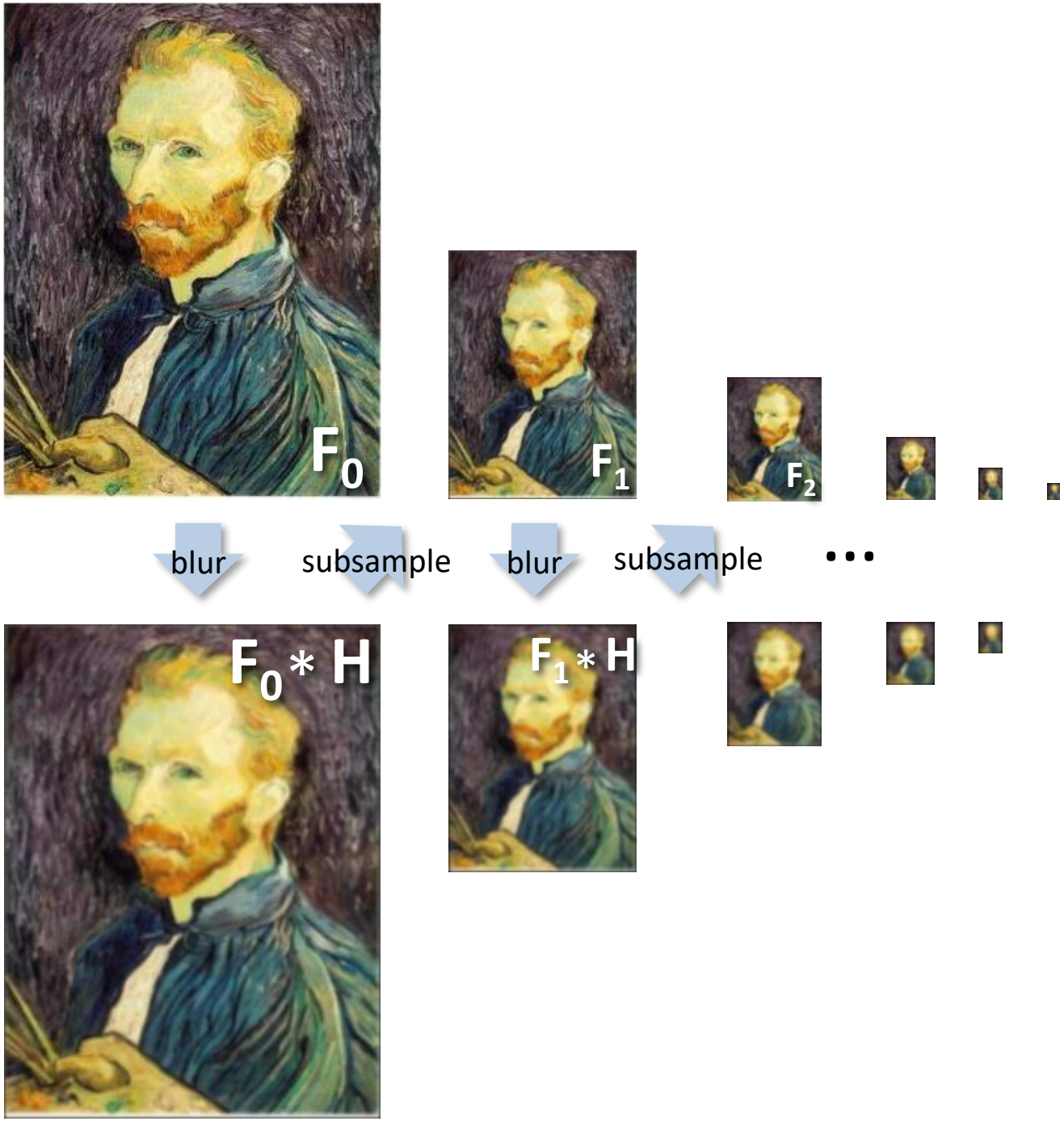
$$\frac{1}{8}\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \qquad \frac{1}{8}\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

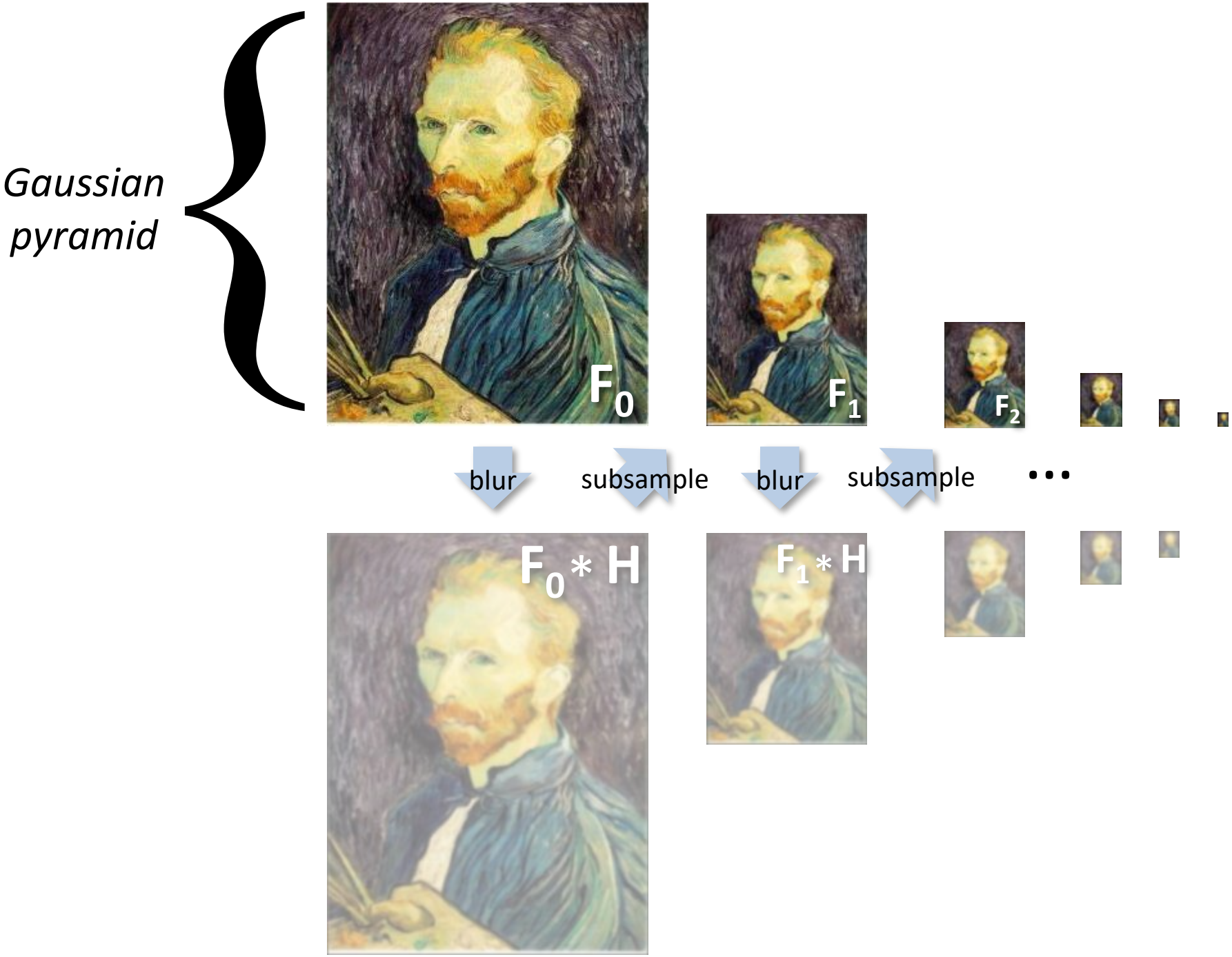
$s_x$ $\qquad$ $s_y$

- The standard defn. of the Sobel operator omits the 1/8 term
  - doesn't make a difference for edge detection
  - the 1/8 term **is** needed to get the right gradient magnitude

# Gaussian pre-filtering

- Solution: filter the image, *then* subsample

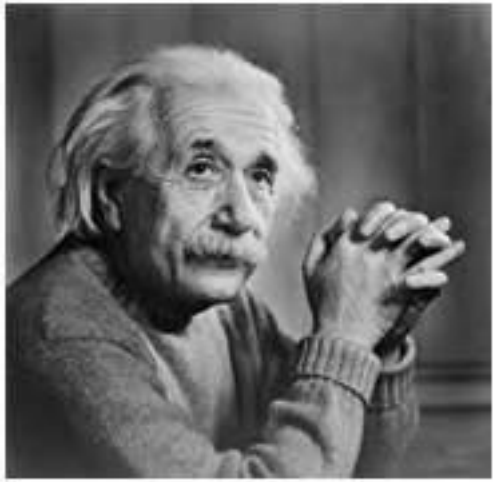


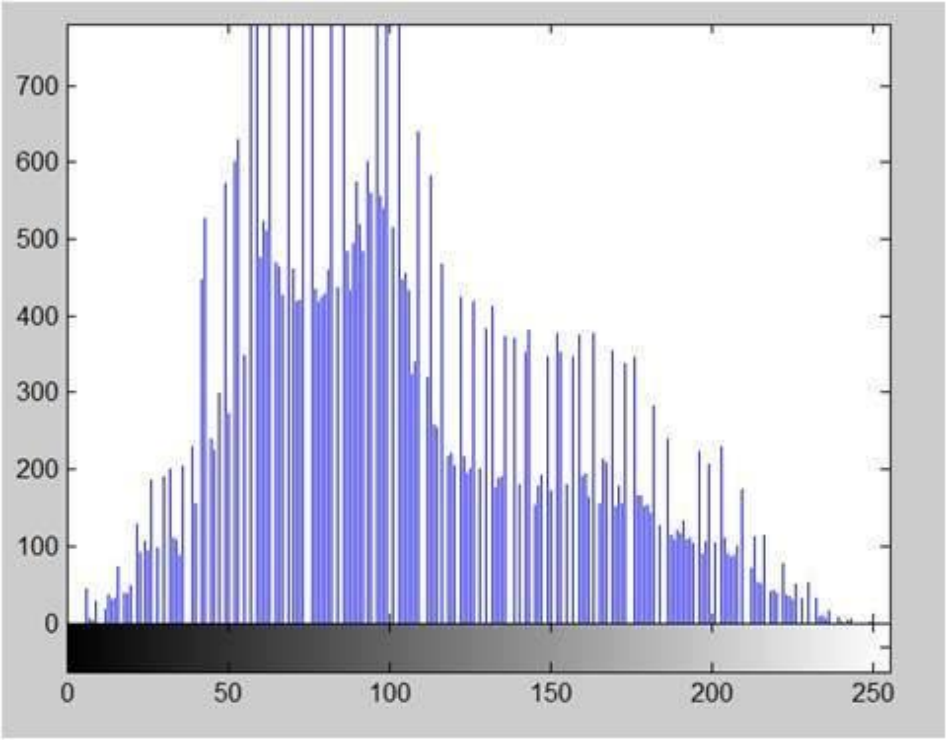

Gaussian pyramid
$F_0$
$F_1$
$F_2$
blur
subsample
blur
subsample
...
$F_0 * H$
$F_1 * H$

# Histograms

- A histogram of a gray-tone image is an array H[*] of bins, one for each gray tone.
- H[i] gives the count of how many pixels of an image have gray tone i.
- P[i] (the normalized histogram) gives the percentage of pixels that have gray tone i.

# Gray image histogram






$$h(i) = \sum_{x} \sum_{y} \begin{cases} 1 & if\ F(x,y) = i \\ 0 & otherwise \end{cases}$$

# Q1

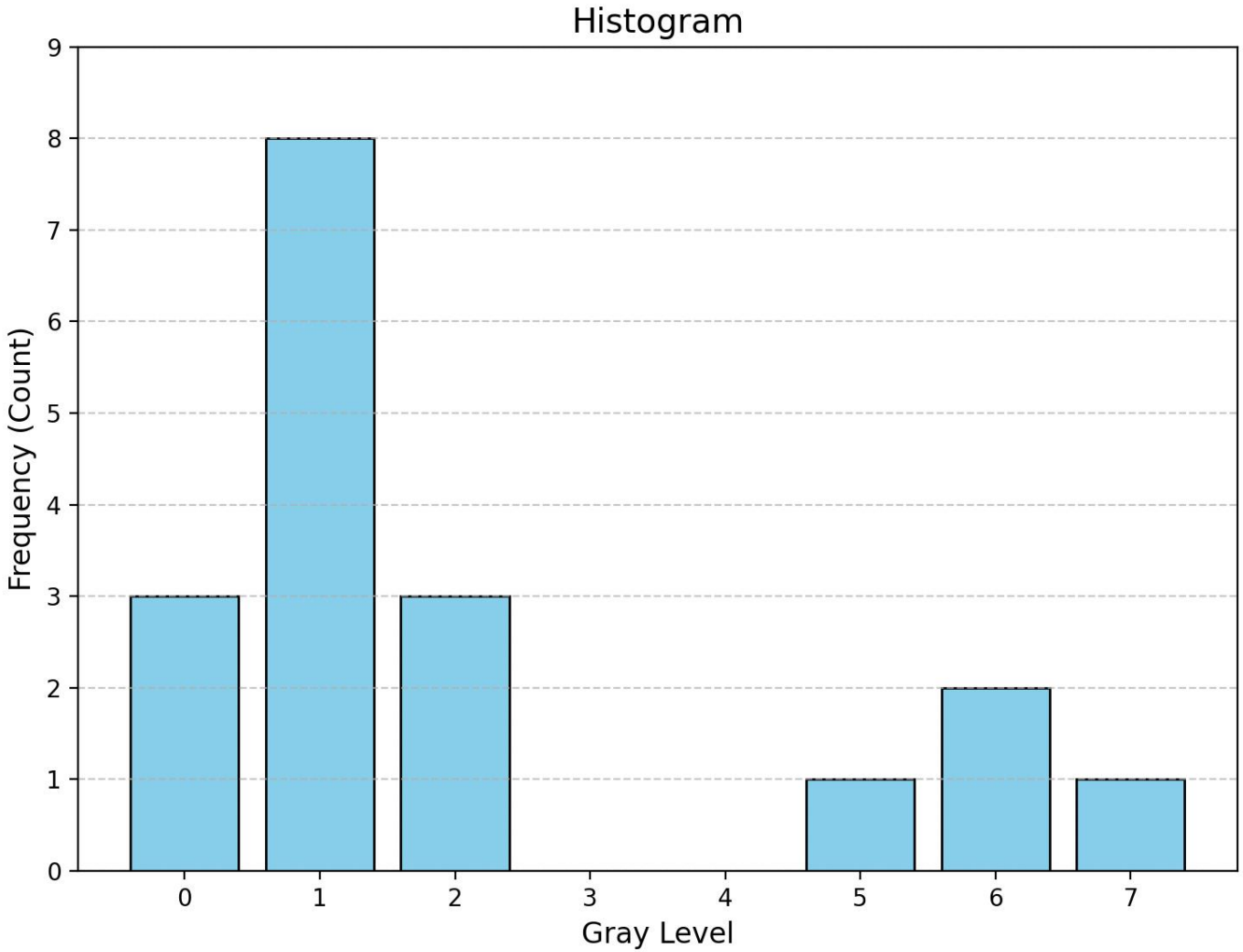
- Consider the image shown below, which has eight possible gray levels. Draw the histograms

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 1 | 1 | 2 | 0 |
| 0 | 1 | 5 | 1 | 0 | 1 |
| 1 | 6 | 7 | 6 | 1 | 2 |

# Q1

Solution:

| Gray level | Coordinates | Count |
|---|---|---|
| 0 | (1,6), (2,1), (2,5) | 3 |
| 1 | (1,1), (1,3), (1,4), (2,2), (2,4), (2,6), (3,1), (3,5) | 8 |
| 2 | (1,2), (1,5), (3,6) | 3 |
| 3 | - | 0 |
| 4 | - | 0 |
| 5 | (2,3) | 1 |
| 6 | (3,2),(3,4) | 2 |
| 7 | (3,3) | 1 |

# Q1

Solution: